Module-V Lattices				
Module - V/ Part - B / 2 Marks				
Sl. No.	**Questions**	**Marks Split-up**	**K – Level**	**CO**
1.	Define partial ordering relation.	2	K1	CO5
2.	Prove that is the set of integers Z the relation R is less than or equal to $\leq$ is a partial ordering.	2	K2	CO5
3.	Define poset with an example.	2	K1	CO5
4.	If a poset has a least element 0, then prove that it is unique.	2	K2	CO5
5.	Define comparable elements.	2	K2	CO5
6.	Define Bounded lattice.	2	K2	CO5
7.	Define lattice homomorphism.	2	K2	CO5
8.	Define lattice isomorphism.	2	K2	CO5
9.	Define Direct product of lattices.	2	K2	CO5
10.	Show that a chain with three or more elements is not complemented.	2	K1	CO5
11.	In a distributive lattice prove that complement of an element, if it exists, is unique.	2	K1	CO5
12.	Show that every interval of a lattice is a sublattice.	2	K2	CO5
13.	In a lattice prove that $a \leq b \Longrightarrow a * b = a$	2	K2	CO5
14.	Draw Hasse diagram of all lattices with upto 5 elements.	2	K2	CO5
15.	Show that every chain is a modular lattice.	2	K2	CO5
16.	Find all the sublattices o the lattice (S_{12}, D) where D is divisibility.	2	K2	CO5
17.	Let $(P, \leq)$ be a poset. If the least element and greatest element exist, prove that they are unique.	2	K2	CO5
18.	Let $(P, \leq)$ be a poset and $a_1, a_2 \in P$, if a_1 and a_2 have a GLB, then prove that it is unique.	2	K2	CO5
19.	Let $(P, \leq)$ be a poset and $a_1, a_2 \in P$, if a_1 and a_2 have a GLB, then prove that it is unique.	2	K2	CO5
20.	Define distributive lattice.	2	K2	CO5

Module - V/ Part - C/ 10 Marks				
Sl. No.	**Questions**	**Marks Split-up**	**K – Level**	**CO**
1.	If R is the relation on the set of integers such that (a,b) ∈ R if and only if $b = a^m$ for some positive integer, show that R is a partial ordering.	10	K3	CO5
2.		10	K3	CO5
3.	Let "A" be a non empty subset of real numbers. Define a relation R on A by xRy if x divides y, for all $x, y \in A$. Show that A is a poset with respect to R.	10	K3	CO5
4.	Let "N" be the set of natural numbers with the relation R as follows. aRb if and only if a divides b . Show that R is a partial order relation on N.	10	K3	CO5
5.	Let $(L, \leq)$ be a lattice with the binary operations meet and join denoted by $*$ and $\oplus$. Then prove that for any a,b,c∈ L the following are true. (i) $a * a = a$ and $a \oplus a = a$ (ii) $a * b = b * a$ and $a \oplus b = b \oplus a$ (iii) $a * (b * c) = (a * b) * c$ and $a \oplus (b \oplus c) = (a \oplus b) \oplus c$ (iv) $a * (b \oplus c) = a$ and $a \oplus (b * c) = a$	10	K3	CO5
6.	Let $(L, \leq)$ be a lattice with the binary operations meet and join denoted by $*$ and $\oplus$. Then prove that a,b∈ L, (i) $a * b$=a if and only if $a \leq b$ (ii) $a \oplus b$=b if and only if $a \leq b$ (iii) $a * b$=a if and only if $a \oplus b$=b	10	K3	CO5
7.	Let $(L, \leq)$ be a lattice. For any a,b,c∈ L the following properties are called isotonicity hold. If $b \leq c$ then prove that (i) $a * b \leq a * c$ and (ii) $a \oplus b \leq a \oplus c$	10	K3	CO5
8.	Let $(L, \leq)$ be a lattice. For any a,b,c∈ L then prove that (i) $a \oplus (b * c) \leq (a \oplus b) * (a \oplus c)$ and (ii) $a * (b \oplus c) \geq (a * b) \oplus (a * c)$	10	K3	CO5

9.	Let $(L, \leq)$ be a lattice. For any a,b,c∈ L then prove that $a \leq c$ if and only if $a \oplus (b * c) \leq (a \oplus b) * c$	10	K3	CO5
10.	Let D_{100}={1,2,4,5,10,20,25,50,100} be the divisors of 100 and let the relation $\leq$ be the relation $a \leq b$ if a/b, then prove that $(D_{100}, /)$ is a poset. Determine (i)GLB{10, 20} (ii)LUB{10, 20} (iii) GLB{5, 10, 20, 25} (iv) LUB{5, 10, 20, 25}	10	K3	CO5
11.	Let D_{30}={1, 2, 3, 5, 6, 10,15, 30} and let the relation D be division on D_{30}. Find (i) all lower bounds of 10 and 15 (ii) all glb of 10 and 15 (iii) all upper bounds of 10 and 15 (iv) the lub of 10 and 15 (v)Draw the Hass diagram for D_{30} with D.	10	K3	CO5
12.	Show that every chain is lattice.	10	K3	CO5
13.	Let $(L, *, \oplus)$ be a lattice defined as an algebraic system with a partial order R on L defined by aRb if and only if $a * b = a$. Show that for any a,b∈ L, GLB$\{a, b\} = a * b$ and LUB$\{a, b\} = a * b$.	10	K3	CO5
14.	Let $(L, *, \oplus)$ is a lattice and let a,b∈ L such that $a \leq b$ then the closed interval $\{a, b\}$ is defined as the set of all x∈ L such that $a \leq x \leq b$. Prove that [a, b] is a sublattice of L.	10	K3	CO5
15.	Let S={a, b, c,d}. Then P(s) is the power set of S consisting of all subsets of S. Then $(P(S), \cap, \cup)$ is a lattice where * is ∩ and ⊕ is ∪. Then prove that $\{\emptyset, \{a\}, \{a, c\}, \{c\}, \{a, b, c\}\}$ is a sublattice.	10	K3	CO5
16.	Prove that every chain is a distributive lattice.	10	K3	CO5
17.	In a distributive lattice $(L, *, \oplus)$ if for any a,b,c∈ L, $a * b = a * c$ and $a \oplus b = a \oplus c$ then prove that b=c.	10	K3	CO5
18.	If S_{42} is the set all divisors of 42 and D is the relation "divisor of" on S_{42}, prove that { S_{42}, D} is a complemented lattice.	10	K3	CO5
19.	If $(L, *, \oplus)$ be a complemented, distributive lattice, then prove that for any a, b∈L, $(a * b)' = a' \oplus b'$ and $(a \oplus b)' = a' * b'$	10	K3	CO5
20.	Show that the axioms of the algebraic lattice $(L, *, \oplus)$ implies a partial ordering on L.	10	K3	CO5